SOFTWARE REQUIREMENTS SPECIFICATION

MediFind

Medicine Availability and Nearby Pharmacy Finder

1. Introduction

1.1 Purpose

MediFind is a web application that helps users search for medicines and find nearby pharmacies where those medicines may be available. This document describes the main requirements of the system.

1.2 Scope

The application allows users to search for medicines, select their current or a manually entered location, and view nearby pharmacies with stock details when available. It also provides login and registration and an admin dashboard for managing medicine, pharmacy, and inventory records.

2. Problem Statement

People may need to visit multiple pharmacies to find a particular medicine because they do not know which nearby pharmacy has it in stock. This can waste time, especially when the medicine is needed urgently. MediFind aims to reduce this difficulty by allowing users to search for a medicine online and view nearby pharmacies with recorded availability, stock quantity, and contact details before travelling.

3. Objectives

- To allow users to search for medicines by name and, where available, generic or brand name.
- To help users find nearby pharmacies based on their selected location and search radius.
- To display pharmacy details and recorded medicine stock status.
- To provide user registration and login.
- To allow administrators to manage medicines, pharmacies, and inventory.
- To store information in a database so updates can be reflected in search results.

4. System Requirements

4.1 Hardware Requirements

- Computer, laptop, tablet, or smartphone.
- Internet connection.

- A modern web browser.

4.2 Software Requirements

- Frontend: HTML, CSS, JavaScript and the frontend framework used in the project.
- Backend: Java Spring Boot REST API.
- Database: MySQL.
- Build tool: Maven.
- Browser: Google Chrome or another modern browser.
- Location services: Browser geolocation and a maps provider when configured.

5. Functional Requirements

- User Registration and Login: Users should be able to register, log in, and log out securely.
- Medicine Search: Users should be able to search for medicines by name and by generic or brand name when those details are available.
- Nearby Pharmacy Search: Users should be able to use their current location or manually select a locality to find nearby pharmacies.
- Search Radius: Users should be able to select a search radius such as 3 km, 5 km, 10 km, or 15 km.
- Availability Details: Search results should show the pharmacy name, address, contact number when verified, distance, stock quantity when known, availability status, and last-updated time when available.
- Stock Status: The system should distinguish available, out-of-stock, and unknown stock. Unknown or outdated stock must not be shown as confirmed availability.
- User Dashboard: Logged-in users should be able to access the user dashboard. Search history may be provided if implemented.
- Admin Dashboard: Administrators should be able to add, view, edit, and deactivate or delete medicine and pharmacy records.
- Inventory Management: Administrators should be able to select a medicine and pharmacy, update stock quantity and status, and save the changes.
- Database Storage: Medicine, pharmacy, user, and inventory information should be stored in the database and remain available after refresh or restart.
- Access Control: Only authorised administrators should be able to perform administrative operations.
- Error Messages: The system should display suitable messages for invalid login, missing search results, input errors, and network problems.

6. Non-Functional Requirements

- Security: Passwords should be securely hashed, and admin functions should be protected.
- Usability: The website should be easy to understand and use.
- Responsiveness: The interface should work on desktop and mobile screens.
- Reliability: The system should handle errors and unavailable data clearly.
- Accuracy: Pharmacy and stock information should not be presented as verified unless supported by reliable records.
- Maintainability: Medicine, pharmacy, and inventory records should be manageable without rebuilding the frontend.
- Performance: Search results should be returned within a reasonable time for the available dataset.
- Privacy: The website should request permission before using the user's current location and should provide manual location selection.

7. Database Requirements

The system uses MySQL to store information. The main tables are:

- Users: Stores user and administrator account details, such as user ID, name, email, password hash, and role.
- Medicines: Stores medicine ID, generic name, brand name, strength, dosage form, category, and verification details where available.
- Pharmacies: Stores pharmacy ID, name, address, locality, city, phone number, coordinates, operating hours, and status where available.
- Inventory: Stores the relationship between a pharmacy and a medicine, including quantity, stock status, and last-updated time.
- Search History (optional): Stores user search terms and search time if the search history feature is implemented.

Each inventory record is linked to one medicine and one pharmacy. A medicine being present in the catalogue does not mean that every pharmacy stocks it. Missing or outdated stock information should be marked as unknown.

8. System Workflow

1. The user opens MediFind and logs in or registers if needed.
2. The user enters a medicine name in the search bar.

3. The user shares their current location or selects a locality manually.
4. The user selects a search radius.
5. The system searches the database and displays matching medicine and pharmacy records.
6. The user checks the pharmacy address, contact details, distance, and recorded stock status before travelling.
7. The administrator logs in separately to add or update medicine, pharmacy, and inventory information.
8. Updated information is saved in the database and appears in subsequent search results.

Note: The availability shown depends on the latest recorded stock information. MediFind should not claim real-time stock unless a live inventory connection is actually available. Medicine prices are not part of the system.